

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / \sqrt{d_k})V \quad (1)$$

Q

$d_{\text{model}} \times d_k$

$$\text{MultiHead}(Q,K,V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O$$

This method improves translation quality across documents.